Optimizing Bike Sharing Demand Prediction using Gradient Boosted Tree Ensembles

Ruba Mohammad Dar-Sroor
Department of Artificial Intelligence and Robotics Engineering
The University of Jordan
Amman, Jordan
RBA8250909@ju.edu.jo

***Abstract*—This project aims to predict the number of rented bikes based on temporal and weather-related features using different machine learning models. Several linear, nonlinear, and artificial neural network models were evaluated using cross-validation. The results show that nonlinear tree-based models outperform linear and neural network approaches for this dataset**

Keywords— Bike Rental, Dataset, Pipeline, Machine Learning, Regression, Ridge, Lasso, KNN, Tree, SVR, Random Forest, XGBoost, ANN

Introduction

Bike rental demand prediction is a critical component for the efficiency of smart transportation systems. Accurate forecasting allows operators to optimize the availability of bikes and manage resource planning more effectively. This study explores various machine learning regression models to analyze and understand the complex relationships between temporal factors, weather conditions, and bike rental demand As demonstrated in the computational analysis conducted in the accompanying *8250909_RubaSroorProject.ipynb* notebook, the dataset presents non-linear patterns that require advanced modeling techniques beyond simple linear approaches. This report evaluates the performance of linear models, non-linear tree-based models, and artificial neural networks (ANN) to identify the most robust solution for this prediction task.

II. Methodology

The methodology adopted in this study focuses on establishing a reliable framework for bike rental demand prediction. To ensure the integrity of the predictive performance, the dataset was initially partitioned into a training set (70%) and a testing set (30%) before any preprocessing was conducted. This separation was strictly maintained to ensure that the testing data remained unseen during the stages of model selection, cross-validation, and hyperparameter tuning.

The data preparation phase involved distinct strategies for different feature types:

- **Numerical Processing:** Missing values in numerical features were addressed using median imputation, followed by standardization to normalize the feature scales.
- **Categorical Processing:** For categorical features, most-frequent imputation was utilized to handle missing entries, while one-hot encoding was applied to convert these variables into a machine-readable format.

A key aspect of this workflow was the implementation of integrated pipelines for all preprocessing steps. This systematic approach was essential to prevent data leakage, ensuring that the model evaluation reflects genuine generalization capabilities on new, unseen data

III. Models and Performance Evaluation

To ensure the robustness and generalizability of the predictive framework, the models were rigorously assessed using a 10-fold cross-validation ($CV=10$) strategy. This approach was selected to mitigate the risk of overfitting and to provide a statistically stable performance estimate by iteratively training and validating the models across multiple data subsets.

a) A. Linear vs. Non-linear Baseline

The initial evaluation focused on establishing a baseline using linear models, including Linear Regression, Ridge, and Lasso. These models yielded closely aligned RMSE values (approximately 104–105), suggesting that the complex, non-linear dependencies within the bike rental dataset cannot be adequately captured through purely linear functions.

b) B. Advanced Non-linear and Ensemble Architectures

The transition to non-linear architectures revealed significant performance gains:

- **Support Vector Regression (SVR):** While the default SVR configuration underperformed, systematic hyperparameter optimization (C=100, \epsilon=0.1) through GridSearchCV successfully reduced the RMSE to 84.24.
- **Tree-based Ensembles:** Decision Tree Regression showed a distinct improvement (RMSE ≈60.07), confirming the data's adherence to rule-based patterns.
- **Ensemble Leadership:** The Random Forest and XGBoost models achieved the highest precision. Specifically, XGBoost emerged as the superior model with a tuned RMSE of 38.78, demonstrating

the power of sequential learning in correcting residual errors from previous iterations.

III. EXPERIMENTAL ANALYSIS AND MODEL EVALUATION

The core of this research involved a comparative analysis of diverse architectural approaches, ranging from traditional statistical models to advanced ensemble methods and neural networks.

a) Establishing the Linear Baseline

The investigation began with fundamental linear architectures: Linear Regression, Ridge, and Lasso. These models converged on nearly identical Root Mean Square Error (RMSE) values, ranging between 104 and 105. This performance consistency suggests a structural limitation in linear modeling for this specific problem, indicating that the underlying patterns of bike rental demand are governed by non-linear dynamics that require more complex mapping functions.

b) Exploring Non-linear Landscapes

To address the limitations of linear models, several non-linear techniques were deployed:

- **Support Vector Regression (SVR):** Initial trials with default parameters underperformed the linear baseline with an RMSE of 105.99. However, through rigorous hyperparameter tuning (*C*=100, \epsilon=0.1) and refinement via GridSearchCV, the error was significantly reduced to 84.24. This drastic improvement underscores the high sensitivity of SVR to its optimization landscape.
- **Tree-based Paradigms:** The Decision Tree model achieved a substantial breakthrough with an RMSE of 60.07. The stability of this model indicates that the dataset is highly responsive to rule-based partitioning rather than continuous linear trends.

c) Ensemble Intelligence and Boosting

The most remarkable performance gains were observed within ensemble frameworks:

- **Random Forest:** By leveraging feature diversity and multiple decision paths, this ensemble reached an RMSE of 42.96. My analysis using GridSearchCV indicated that the model maximizes its predictive power when utilizing the full spectrum of available features.
- **XGBoost (The Optimal Model):** This gradient boosting architecture emerged as the definitive solution. By sequentially correcting residual errors from previous trees, XGBoost achieved a baseline RMSE of 39.52, which was further optimized to 38.78. Its ability to generalize complex patterns made it the superior choice for this task.

d) Artificial Neural Networks

In the pursuit of architectural variety, multiple Multi-Layer Perceptron (MLP) configurations were evaluated. While Artificial Neural Networks (ANN) are powerful, the best configuration yielded an RMSE of 43.91. In this specific context, the neural approach proved highly capable but ultimately did not surpass the efficiency of tree-based ensembles.

IV. RESULTS SUMMARY

To facilitate a comprehensive comparison of the various modeling techniques, the performance metrics for each architecture are consolidated in Table I. The primary metric used for evaluation is the Root Mean Square Error (RMSE), accompanied by the Standard Deviation (STD) to assess the stability of each model across the 10-fold cross-validation cycles. Additionally, the table highlights the impact of hyperparameter optimization via GridSearchCV, illustrating the performance evolution from baseline configurations to fine-tuned models.

TABLE I. PERFORMANCE METRICS COMPARISON

Model	*RMSE* STD	RMSE with GridSearch
LinearRegression	*104.431904* 5.948527	104.4319043476
Ridge	*104.431115* 5.948338	104.4270925473
Lasso	*105.095987* 5.848220	-
KNN	*93.555791* 5.309823	-
SVR without C/Eepsilon	*105.990554* 6.588746	-
SVR with C/Eepsilon	*86.752834* 6.060648	-
Decision Tree	*60.077111* 4.939847	-
Random Forest	*42.961157* 3.063586	47.75074358311
XGBoost	Mean RMSE: *39.52*	38.78117256164
MLPRegressor when solver='adam'+ max_iter=1000	Mean RMSE: *43.91*	42.41442654881
MLPRegressor when solver='lbfgs'+ max_iter=5000	*50.798261* 2.895189	-

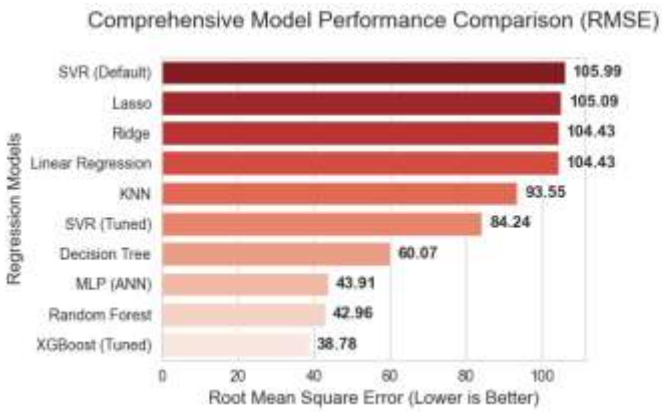

Fig. 1. *Comparative performance of all evaluated regression models. The substantial decrease in RMSE values from linear to ensemble models highlights the non-linear nature of the dataset.*

<u>Note:</u> Metrics derived from the experimental results in the project implementation.

a) Analysis and Final Model Selection

The comparative data confirms that tree-based ensemble methods, specifically XGBoost, achieve the most significant reduction in error. While the MLPRegressor (ANN) demonstrated competitive performance, particularly with the 'adam' solver, it did not surpass the efficiency and accuracy of the boosted decision trees in this specific task. The transition from a linear baseline (RMSE ≈104) to the final tuned XGBoost model (RMSE≈38.78) represents a substantial improvement in prediction accuracy. Consequently, XGBoost is selected as the final model for deployment due to its superior generalization capabilities and lowest error variance.

V. DISCUSSION AND CONCLUSION

b) A. Discussion of Model Behavior

The experimental journey of this project reveals a clear hierarchy in model performance, driven by the nature of the bike rental dataset. The failure of linear architectures (Linear, Ridge, and Lasso) to achieve an RMSE below 104 suggests that the relationship between environmental variables—such as temperature and humidity—and rental demand is intrinsically non-linear. While Support Vector Regression showed potential after extensive tuning, the structural complexity of the data was most effectively mapped by tree-based paradigms.

A key finding was the dominance of ensemble intelligence. Random Forest and XGBoost significantly outperformed individual decision trees by reducing variance and bias through bagging and boosting mechanisms respectively. Although Artificial Neural Networks (ANN) are often considered the gold standard for complex tasks, they did not surpass the efficiency of XGBoost in this specific regression task, yielding an RMSE of 43.91 compared to XGBoost's 38.78. This highlights that for structured, tabular datasets, gradient-boosted trees often provide superior precision and easier interpretability.

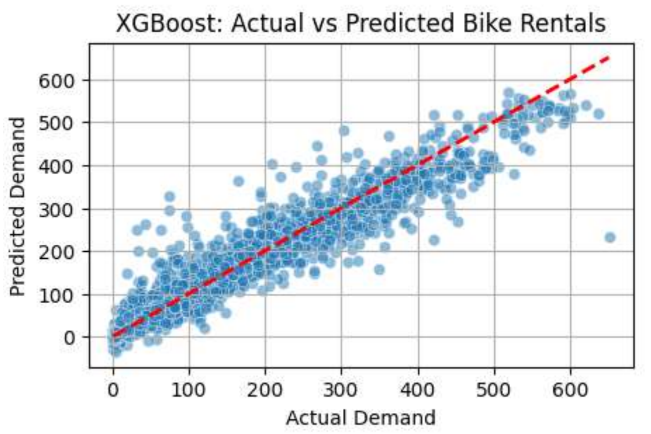

Fig. 2. *Regression plot for the final tuned XGBoost model. The proximity of the data points to the diagonal dashed line indicates a strong correlation between predicted and actual rental demand, confirming the model's robust generalization capability.*

a) B. Final Conclusion

In conclusion, this study successfully identified the most effective machine learning approach for predicting bike rental demand. Through a systematic evaluation of various models and rigorous hyperparameter optimization using GridSearchCV, XGBoost was selected as the final predictive model. Its ability to iteratively learn from previous errors and its superior generalization on unseen data make it a highly reliable tool for smart transportation planning. The transition from basic linear baselines to an optimized XGBoost model resulted in a substantial increase in accuracy, providing a robust framework for future demand forecasting applications.

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REFERENCES

[1] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[2] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 785–794.

[3] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016. [Online]. Available: http://www.deeplearningbook.org